

Arun Jayaraman

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Summary

Senior Scientist and translational data scientist with 6+ years of applied computational research across pharmaceutical R&D and academia, specializing in immunology, single-cell and multimodal omics, biomarker strategy, and precision medicine. Leading computational strategy for immunology clinical development and cross-indication programs, translating complex molecular and clinical data into actionable investment and portfolio decisions. Developing AI-enabled analytical workflows and tools that integrate validated internal omics data with external biological evidence to accelerate translational insight and indication prioritization.

Work Experience

Senior Scientist

AstraZeneca

Jan 2023 – Present

Gaithersburg, MD

- Primary translational data scientist for AZ's early and clinical-stage RA programs, leading biomarker strategy and multi-omics analyses supporting translational, clinical development, and investment decisions.
- Lead computational strategy across academic and consortium collaborations spanning pre-RA through established autoimmune disease, integrating longitudinal multi-omic, single-cell, spatial, immune-repertoire, and clinical data to inform disease positioning and biomarker development.
- Develop advanced single-cell and multimodal analyses across internal and public datasets, including a scVI-based cross-disease T-cell atlas, matched blood/tissue studies, and integrated CITE-seq analyses.
- Influence portfolio strategy across RA, pediatric JIA, cell therapy, TCEs, and multispecific programs through indication assessments, biomarker analyses, competitive positioning, and synthesis of molecular and clinical evidence.
- Evaluate emerging biomarker technologies and guide assay selection, experimental design, cohort construction, and validation; benchmarked three autoantibody-discovery platforms and provided recommendations for further investment based on sensitivity and technical maturity.
- Co-lead development of AI-enabled workflows over internal omics repositories to combine validated internal data with external evidence and accelerate cross-indication assessment and indication prioritization.
- Served as project lead for a centralized R Shiny translational-data platform enabling self-service molecular and signature analyses; coordinated collaborative development and production deployment using GitHub and CI/CD.

Associate Scientist II

AbbVie Genomics Research Center

Jan 2022 – Jan 2023

North Chicago, IL

- Identified disease drivers in therapeutic non-responders through integrated single-cell and bulk transcriptomics, including a pathogenic synovial fibroblast population enriched in RA non-responders.
- Applied longitudinal mixed-effects modeling to upadacitinib-treated arthritis cohorts, validating drug-associated molecular signatures and informing therapeutic strategy.

Graduate Researcher

Emory University, The Davis Lab

Sept 2020 – Dec 2021

Atlanta, GA

- Led scRNA-seq studies of cardiac stromal/progenitor populations and collaborated with Yale School of Medicine on T-cell responses associated with cardiac allograft rejection, contributing to multiple peer-reviewed publications.

AI/ML-Driven Drug Discovery and Cheminformatics

AbbVie Innovation Center

Summer 2020

Urbana, IL

- Applied generative models (VAE, GAN, RL) to *de novo* molecular design and developed a hierarchical neural network to predict chemical reaction conditions for drug-discovery workflows.

Education

Professional Certificate in Artificial Intelligence

Stanford School of Engineering, Artificial Intelligence Professional Program

Expected Oct 2026

Online

M.S., Biomedical Engineering

Georgia Institute of Technology

Dec 2021

Atlanta, GA

B.S., Chemical Engineering, With Distinction (Dean's List)

University of Illinois Urbana-Champaign

May 2020

Urbana, IL

Technical Skills

- **Single-cell & multimodal:** scRNA-seq, CITE-seq, scVI, Seurat, Scanpy, Harmony, CCA, pseudobulk differential expression and abundance, Monocle 3, AUCell
- **Repertoire, cytometry & spatial:** single-cell and bulk BCR/TCR-seq (Cell Ranger VDJ, scRepertoire); CyTOF; Xenium, GeoMx, Space Ranger
- **Bulk omics & integration:** DESeq2, edgeR, limma, mixed-effects models (lme4, dream), MOFA+, WGCNA, GSVA, singscore; Olink proteomics; STAR, Salmon
- **Machine learning & AI:** PyTorch, TensorFlow, scikit-learn; deep generative models; reinforcement learning; AI agents; AI coding assistants
- **Programming & infrastructure:** Python, R, Bash, SQL, JavaScript; Git/GitHub, CI/CD; Linux HPC, AWS; Nextflow, Snakemake; R Shiny, ggplot2, Matplotlib

Selected Publications, Patents & Presentations

Patent

- "PAD ENZYMES AS BIOMARKERS" (2025). U.S. Provisional Patent Application No. 63/811,164.

Peer-reviewed publications (* co-first author)

- Korutla L, Hoffman JR, ..., **Jayaraman A**, et al. (2024). Circulating T cell specific extracellular vesicle profiles in cardiac allograft acute cellular rejection. *Am. J. Transplant.*
- Hoffman JR*, **Jayaraman AR***, Bheri S, Davis ME (2022). Single-cell RNA sequencing reveals distinct cardiac-derived stromal cell subpopulations. *J. Cardiovasc. Dev. Dis.* 9:374.
- Gunasekaran M, Mishra R, Saha P, Morales D, Cheng W-C, **Jayaraman AR**, et al. (2022). Comparative efficacy and mechanism of action of cardiac progenitor cells after cardiac injury. *iScience* 25:104656.
- Hoffman JR, Park H-J, Bheri S, **Jayaraman AR**, Davis ME (2022). Comparative computational RNA analysis of cardiac-derived progenitor cells and their extracellular vesicles. *Genomics* 114:110349.
- Park J, **Jayaraman A**, Schrader AW, Hwang GW, Han H (2020). Controllable modulation of precursor reactivity for synthesis of high-quality quantum dots. *Nat. Commun.*

Conference presentations

- Csomor E, **Jayaraman A**, Stewart L, ..., McInnes IB, Goodyear CS (2025). Prevalence of IFN-high rheumatoid arthritis in early and established RA patients. EULAR 2025 (ABS0252). *Ann. Rheum. Dis.* 84(Suppl. 1):1950–1951.
- Sharrack S, Yu X, **Jayaraman A**, ..., Mankia K (2025). Elevated serum peptidylarginine deiminase 4 (PAD4) levels in anti-CCP antibody positive at-risk individuals with arthralgia who progress to rheumatoid arthritis. Posters, EULAR 2025 (POS0067; *Ann. Rheum. Dis.* 84(Suppl. 1)) and ACR Convergence 2025.
- Azim A, **Jayaraman A**, et al. (2024). Nasal sampling identifies disease-relevant signals in a real-world airways disease cohort. Oral presentation, ERS Congress 2024. *Eur. Respir. J.* 64(Suppl. 68):OA3764.
- Faner R, Azim A, **Jayaraman A**, et al. (2024). The NOVELTY nasal transcriptome identifies mechanisms in asthma and COPD exacerbators. Poster, ERS Congress 2024. *Eur. Respir. J.* 64(Suppl. 68).
- van den Berge M, ..., Azim A, **Jayaraman A**, et al. (2024). The nasal transcriptome reflects clinically relevant disease activity in NOVELTY, with a different signal in asthma and in COPD. Poster, ERS Congress 2024. *Eur. Respir. J.* 64(Suppl. 68).